

Cover Letter

Dear Editor,

I would like to submit a manuscript titled "PACE: Probabilistic Adaptive Contention Control for Non-Primary Channel Access in IEEE 802.11bn" by Taewon Song and Yeunwoong Kyung for consideration in the Journal of Network and Computer Applications.

In this manuscript, we propose PACE, a distributed probabilistic contention control scheme for non-primary channel access (NPCA) that lets stations track the time-varying fair-share transmission probability of a depleting transmission window without central coordination. The core problem is that the IEEE 802.11bn draft has stations reuse binary exponential backoff on the non-primary channel, yet that rule was designed for open-ended contention: within a bounded window, the doubled backoff frequently exceeds the time remaining, so stations fall silent while the channel idles. PACE addresses this by combining a deadline-scaled initialization, a multiplicative feedback rule in which stations copy the successful sender's probability on solo receptions, and a self-exclusion rule that silences stations whose frame no longer fits the remaining window. Under IEEE 802.11 standard timing on a channel shared with native stations, PACE keeps the total useful airtime within 3% of a genie-aided fair-share reference under RTS/CTS-protected access, exceeds the standard-compliant NPCA baseline by up to 63% there and by up to 4.3 times under basic access, and holds the visiting stations' airtime at or below their population-proportional share. An ablation study confirms that both mechanisms are necessary for these gains.

This work directly addresses the Journal of Network and Computer Applications' focus on wireless networks and emerging standards for network protocols: the protocol contribution lies in adapting distributed probability adaptation to the finite-window contention regime that IEEE 802.11bn introduces, and the engineering application is standard-compatible channel access for next-generation Wi-Fi in dense overlapping-BSS deployments.

Both authors have reviewed and approved the manuscript. This manuscript has not been published in any journal and is not under consideration for publication elsewhere.

Sincerely yours,

Yeunwoong Kyung

Department of Electronic Engineering, Seoul National University of Science and Technology

ywkyung@seoultech.ac.kr